SPURTHI PATTANASHETTI

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PROFILE

AI Engineer with hands on experience in machine learning, time series analysis and backend API development. Skilled in building and deploying data driven solutions, with focus on real time prediction and scalable applications.

EDUCATION & CERTIFICATIONS

Bachelor of Engineering — Electronics & Communication *2025*

KLE Technological University

AI Engineering Certification *(In Progress) — DataMites Institute*

EXPERIENCE

AI/IOT Developer Intern - Autoretro Tech Pvt Ltd *Feb 2025 – May 2025*

- Integrated Generative AI (Google Gemini API) into production IoT workflows for intelligent automation and real-time home device control.
- Developed cross-platform mobile application using Flutter with REST API integration connecting mobile interface, IoT hardware (ESP32).
- Built and tested application features in Android Studio, implementing natural language-based device control through Gemini API integration.
- Improved component reusability using modular UI architecture in Flutter for scalable feature deployment.

Tech Stack: Flutter, Dart, Google Gemini API, Android Studio, ESP32, Arduino IDE, REST APIs

PROJECTS

UAV Ground Object Detection & Multi-Target Tracking System [GitHub]

- Built real-time UAV surveillance system using YOLOv8 and custom Centroid Tracker, achieving multi-object detection and persistent ID tracking at 20+ FPS on CPU with confidence-threshold filtering.
- Developed GCS-style Tkinter desktop dashboard with live camera feed, bounding box visualization, motion trail rendering per tracked object, and real-time stats panel showing FPS, detection count, and active tracks.
- Engineered Euclidean distance-based centroid matching with configurable disappearance tolerance and frame-skip optimization, doubling effective FPS while maintaining tracking continuity across frames.

Technologies: Python, OpenCV, YOLOv8, NumPy, Tkinter, Pillow, Real-time Video Processing

Smart Ambulance Vital Monitoring & Early Warning System [GitHub]

- Built ML-powered early warning system achieving 98.9% recall and 81.1% precision, detecting patient deterioration 5–19 minutes in advance using time-series anomaly detection.
- Engineered 20+ clinical features (Shock Index, MAP, rolling statistics) and implemented multi-stage artifact filtering pipeline ensuring high-quality validated inputs.
- Deployed production-grade FastAPI model and reduced false alarms by 45% via hyperparameter tuning while maintaining safety-critical recall.

Technologies: Python, scikit-learn, pandas, FastAPI, Isolation Forest, Feature Engineering, Time-Series Analysis

TECHNICAL SKILLS

Languages: Python, C++, SQL

Machine Learning & Statistical Modeling: Time Series Forecasting, Statistical Modeling, Random Forest, XGBoost, SVM
Deep Learning & Computer Vision: CNN, RNN, PyTorch, OpenCV, YOLOv8, Object detection, Multi-Object Tracking
Backend & APIs: FastAPI, REST API, ML Model Deployment, Basic Django

Tools & Technologies: Docker, Git, Flutter, ESP32

Data Analysis & Visualization: Pandas, NumPy, Matplotlib